Energy-Efficient Paging for Duty-Cycled LTE Backscatter

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***Abstract*—Benefiting from the continuous and ubiquitous transmission from LTE base stations (eNodeBs), LTE backscatter is more efficient and reliable than traditional backscatter operating in ISM bands. However, ISM bands can utilize intermittent signals to synchronize low-duty cycle backscatter and receivers, a mechanism not available in LTE due to its continuous transmission. Consequently, LTE backscatter and receivers (UE) may not have overlapping operational periods, making it possible that the UE might never receive the backscattered data. To solve this issue, we use LTE-standard paging, one of the LTE power management mechanisms, to wake up the receiver and allow it and the tag to maintain a consistent sleep-wake schedule. Following LTE messages, our tag employs a new low-power state machine to accommodate different phases of the UE. We prototype PScatter using FPGAs, SDR eNodeBs, and UEs. Extensive results show that PScatter supports the paging mechanism similarly to active LTE devices and achieves better energy efficiency and more effective transmission than prior LTE backscatter. It consumes 21x lower power than active LTE radios for paging decoding. For end-to-end, PScatter achieves more than 6.7x better energy efficiency than previously reported LScatter+. Moreover, PScatter delivers an effective transmission ratio of up to 100%.**

I. Introduction

Enabling ubiquitous connectivity for everything in the physical world has been the long-term vision of Internet-of-Things (IoT), where tiny devices digitize the real world through pervasive sensing [1]–[3], intelligent interaction [4], [5], and mobile control [6], [7]. Backscatter promises to be a desirable alternative to active radios since it is inherently ultra-low-power without carrier generation [8], [9]. Consequently, backscatter radios consume three to four orders of magnitude lower power than active radios. Especially ambient backscatter systems that reuse existing infrastructure have received considerable attention, unlike RFID systems that require dedicated and expensive readers [10]–[13]. Nevertheless, Bluetooth [14], [15], WiFi [16], [17], ZigBee [18], and LoRa [19], [20] backscatter radios have several limitations. First, the transmissions of these signals are bursty and intermittent, which makes it impossible to provide stable carriers. Second, these signals have limited coverage. Third, these signals share the crowded ISM band and suffer from interference from unintended carriers. In contrast, cellular signals have continuous transmission, ubiquitous coverage, and diverse frequencies, which make them advantageous over WiFi and other signals for backscatter [21]–[23].

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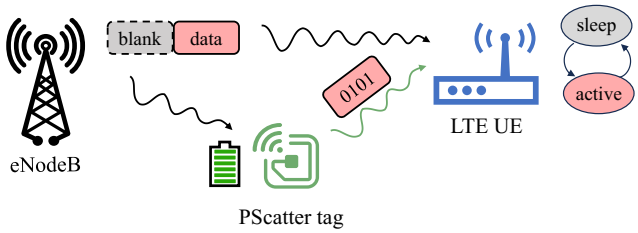

Fig. 1: Conceptual design of PScatter. Without affecting the original LTE transmission, PScatter allows tags and UEs to maintain consistent sleep and wake-up schedules for energy-saving and reliable transmission.

However, traditional cellular backscatter systems operate under the assumption that the User Equipment (UE) is constantly monitoring the backscatter channel. State-of-the-art LTE backscatter systems, such as LScatter, can backscatter signals as long as sufficient power is available [24], [25]. Furthermore, the LTE signal is continuous, allowing the backscatter device to reflect the signal at any time, which exacerbates energy consumption. Hence, LScatter [24] has to transmit data at a low-duty cycle in alternating charge-backscatter mode. But in practice, in both the ISM band and cellular band, devices typically operate in low-duty cycle mode to conserve energy due to intermittent network traffic [26]–[28]. For example, the device wakes up when there is traffic on the network and goes to sleep when there is no traffic. In the ISM band, the receiver has the autonomy to decide when to wake up and check for incoming data. This flexibility allows it to align with the backscatter device's transmission schedule. As a result, traditional backscatter systems can reflect signals at any time, provided it has sufficient energy, as the receiver is ready to receive data [14], [18], [29]. However, cellular bands differ from ISM bands in that who decides when the receiver wakes up. In the cellular band, the UE does not have the same level of control over its sleep and wake schedule. The cellular network determines the paging cycle, i.e., the intervals at which the UE wakes up to check for incoming data [23], [30]. Consequently, if the UE's sleep-wake schedule does not align with the backscatter device's transmission schedule, the UE may never receive the backscattered data.

Such a difference also determines that the ISM wake-

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up mode is not applicable to LTE networks. For example, SyncScatter [31] designs a simple Packet Length Modulation (PLM) wake-up packet that indicates incoming WiFi packets to avoid the tag being in the high-power synchronization stage for a long time. This wake pattern matches WiFi perfectly. However, if it is used for LTE backscatter, SyncScatter's wake-up makes the UE and the backscatter device miss the consistent low-duty cycle transmission mode since the UE follows paging with Discontinuous Reception (DRX), resulting in backscatter failure. To ensure energy-efficient and reliable transmission for LTE, the backscatter transmission must follow the LTE protocol. Legacy LTE supports paging mode with DRX to enable UE to periodically wake up and monitor paging messages at specific subframes [21], [32]. Unfortunately, active LTE paging does not work directly with LTE backscatter. The reason is that it requires the receiver to incorporate a high-frequency oscillator to correlate with the received LTE signal to locate a specific subframe, which inevitably incurs high power consumption that is undesirable for energy-constrained backscatter devices.

To address the above issues, we present PScatter, the first LTE backscatter system that utilizes a standard paging mechanism for energy saving and reliable transmission. As shown in Fig. 1, based on the standard paging mode, a fundamental part for LTE, 5G, and likely future 6G networks as well, PScatter allows tag and UE to maintain a consistent sleep and wake schedule, avoiding wasted energy and invalid transmissions. In addition, PScatter does not change the original duty-cycled LTE transmission to support energy-efficient active and backscatter communications. To achieve PScatter in practice, however, we need to address two main challenges.

Challenge 1: How to design a state machine to handle different LTE transmission phases?

In a typical LTE paging cycle, there are active and sleep phases. However, no prior LTE backscatter systems are aware of the change in those phases. For example, LScatter [24] tag is always active at high power as long as it has enough energy for backscatter and its energy is quickly depleted. Although SyncScatter [31] controls the tag using the duty-cycled wake-up mode to prevent it from being in the high-power synchronization stage for a long time, it cannot determine the wake-up moment of the tag due to incompatibility with LTE. As a result, it cannot achieve effective energy saving for LTE backscatter. To solve this, we design a low-power state machine on the backscatter tag, which is fully compatible with LTE paging cycles and achieves high energy efficiency.

Challenge 2: How to demodulate complex paging messages on tags?

Standard paging is necessary to ensure that tags and UEs maintain consistent sleep and wake schedules. However, there are some difficulties in implementing the paging process on the tag. First, the base station continues to transmit LTE signals, and the tag cannot identify the paging signal and determine the presence of a UE. Second, paging signal processing requires energy-intensive digitization and signal processing that cannot be accommodated on backscatter devices. Although standard

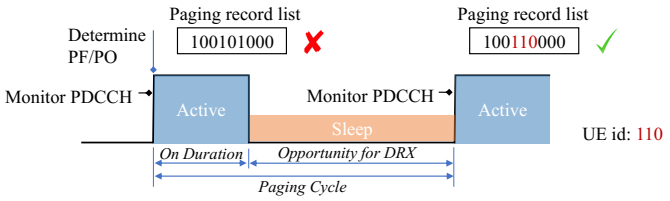

Fig. 2: Paging reception process in LTE.

UEs can demodulate paging messages, they cannot communicate with backscatter devices. Our solution is that the UE forwards the paging message to the tag in an Amplitude Modulation (AM) way to inform the existence and transmission schedule of the UE. As a result, the tag and the UE maintain a consistent transmission schedule following the LTE protocol, and the UE can receive backscattered data efficiently.

We prototype PScatter using customized tags implemented by FPGAs and LTE eNodeB/UE realized by USRPs. Through extensive empirical evaluation, we show that

- To transmit 1 Kbit tag data for the end-to-end transmission, the energy efficiency of PScatter is 0.16 μJ/bit, which is 6.7x lower than that of LScatter+, while LScatter suffers from transmission failures.
- For a single paging cycle of 64 radio frames (RFs), the power of PScatter is 3.01 mW, which is 21.0x, 3.0x, and 3.0x lower than active LTE radios, LScatter, and SyncScatter. Specifically, PScatter has four main states, including static (0.9 mW), wake-up (3.9 mW), active (9.0 mW), and sleep (3.0 mW) states.
- PScatter achieves up to 100% paging identification rates across different transmission power and distance settings. Specifically, with transmission power greater than 0 dBm, the paging identification rate reaches 95%, and it remains above 90% at distances from 1 m to 11 m.

II. Motivation

A. Paging Procedure in LTE Networks

To reduce power consumption, LTE introduces a DRX mechanism [21], [32]. A typical DRX cycle includes On Duration in the active state and Opportunity for DRX in the sleep state. The DRX mechanism comprises two modes: connected mode DRX (CDRX), exclusive to the RRC_CONNECTED (Radio Resource Control) state, and a paging process [23], [30], applicable to UEs in both RRC_IDLE and the RRC_CONNECTED states. Only paging is discussed here. The paging process is that the network side sends paging messages, such as incoming calls and system information changes, to the UE at a specific time. The process of paging reception is shown in Fig. 2. First, the UE needs to determine the moment of paging, which is identified by the Paging Frame (PF) and Paging Occasion (PO). One PO is a subframe where there may be a Paging Radio Network Temporary Identifier (P-RNTI)

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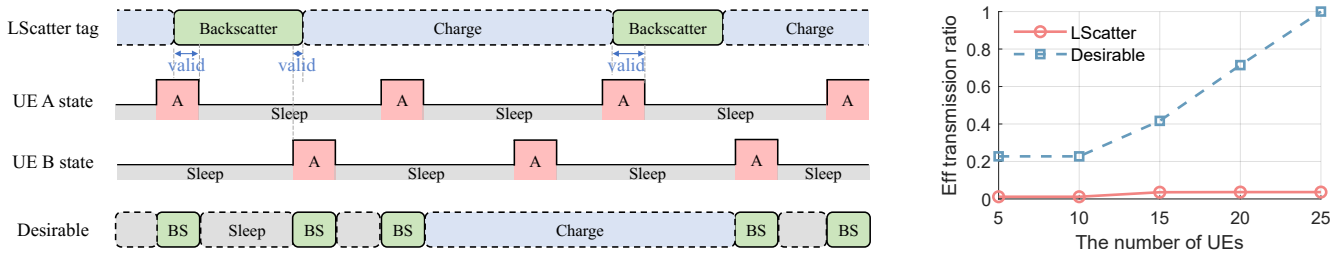

(a) Low duty cycled LTE backscatter transmission with multi-UE.

(b) Effective transmission ratio.

Fig. 3: Performances of LTE backscatter transmission with duty-cycled UEs. Since LScatter has no paging, the transmission schedules of tag and UEs are inconsistent, and many transmissions are invalid, resulting in a low effective transmission ratio. However, each transmission of the desirable LTE backscatter with paging is valid.

transmitted on a Physical Downlink Control Channel (PDCCH) addressing the paging message. One PF is one radio frame, which may contain one or multiple Paging Occasion(s). When DRX is used, the UE only needs to monitor one PO per DRX cycle. PF and PO are determined based on the DRX parameters provided in the system information. PF is given by the following equation: $SFN \bmod T = (T/N) * (UE_ID \bmod N)$ where T is the DRX cycle of the UE, $nB \in \{4T, 2T, T, T/2, T/4, T/8, T/16, T/32\}$, $N = \min(T, nB)$. Index i_s pointing to PO from the subframe pattern will be derived from the following calculation:$i_s = \lfloor UE_ID/N \rfloor \bmod Ns$ where $Ns = \max(1, nB/T)$, $UE_ID = IMSI \bmod 1024$. Second, in each paging cycle, the UE wakes up at the PF and PO and monitors the PDCCH subframe scrambled by the P-RNTI to determine whether a paging message exists. Finally, if the PDCCH indicates that a paging message is transmitted in the subframe, the UE reads each PagingRecord in the PagingRecordList to confirm whether its UE identity matches a certain *ue-Identity*. In summary, unlike continuous LTE signals, paging messages are transmitted intermittently. These messages carry service data, such as incoming calls, at specific paging intervals based on network traffic conditions.

B. How Prior Systems Behave with Paging

Fig. 3a shows the performance of LTE backscatter transmission with paging. LScatter [24] only considers signal synchronization and backscatter modulation and does not realize the importance of paging. Therefore, even if the LTE UE is in a dormant state that cannot receive backscattered signals, the LScatter tag is still active and continuously reflects signals, which causes energy waste and backscatter transmission failure. For reference, Apollo [33] takes 2.64 s to fully charge 1000 μF capacitor using 6 dBm RF energy. The maximum time for this capacitor energy to be used for LTE backscatter modulation is $1000\mu F * 4.1V/14.85mW = 276ms$. As a result, LScatter transmits data in a charge-backscatter manner with a low-duty cycle. Since LScatter has no paging, it cannot be consistent with the UE's sleep and wake-up schedule. It is possible that when UEs are sleeping, the tag transmits data, and when UEs wake up, the tag is charging. We evaluate the performance of the LScatter tag communicating with multiple

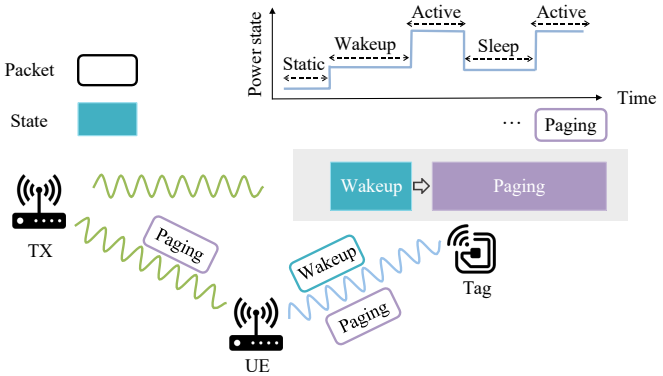

Fig. 4: System overview of PScatter.

UEs within the duty cycle. We set the paging cycle to 640 ms and the active duration to 20 ms. To transmit 100 Bytes tag data, Fig. 3b shows that when there are 20 UEs, the minimum transmission delay of LScatter is 2740 ms, and its effective transmission ratio (ratio of effective transmission time to total transmission time) is $100/2740 = 3.6\%$, while the minimum desirable delay is 100 ms and its effective transmission ratio is 100%. Because LScatter has no paging and transmits data at a low-duty cycle, many transmissions are invalid, which results in an extremely low effective transmission ratio. SyncScatter [31] is different from LScatter in that it designs a wake-up mechanism to switch from a low-power state to a high-power state. Due to incompatibility with LTE, SyncScatter cannot determine the paging frame and paging occasion without paging mode, which makes it randomly enter the active state from the wake-up state. It is possible for the tag to remain active to reflect signals when the LTE UE is dormant. In conclusion, prior backscatter works are incompatible with paging, resulting in wasted energy and ineffective transmission.

III. PSCATTER DESIGN

A. Overview

Fig. 4 shows the overview of PScatter. Once the tag detects a wake-up packet, it enters a low-power wake-up state until it is woken up to a high-power active state, which indicates

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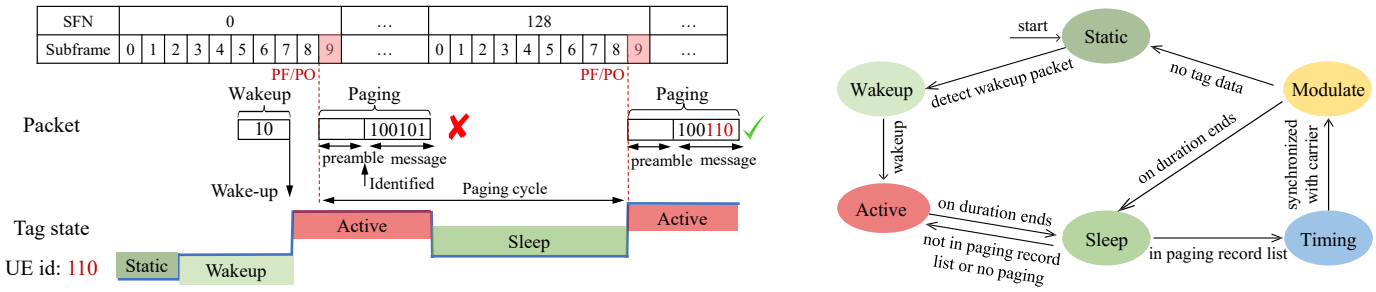

(a) Paging mechanism of PScatter.

(b) State switching.

Fig. 5: State switching of PScatter. (a) When the tag detects a wake-up packet, it switches from the static state to the wake-up state to decode the wake-up packet. Once the tag is awake, the tag enters the active state, identifies the paging packet to determine the PF and PO, and decodes the packet. Since the UE ID is 110, the ID does not exist in the message, and the tag enters the sleep state until it matches the ID. (b) The state switching of the PScatter tag.

that a paging message is imminent. To enable the energy-constrained tag to decode the paging message, the UE forwards the paging message to the tag using AM method. The tag in the active state decodes the forwarded paging packet, enters the periodic paging mode, and switches between active and sleep states until it finds the UE ID in this packet. Once the ID is found, the UE will be ready to receive the backscattered signal. Then, the tag starts to backscatter the LTE signal from the eNB (Evolved NodeB). Such a duty-cycled paging mechanism achieves high energy efficiency and effective transmission by switching among different power states.

B. State Machine

Typically, receivers switch between different states using low-duty cycled transmission modes to save power according to network traffic. To ensure efficient and reliable backscatter transmission, tags often need to maintain a state-switching table that aligns with the state of the receiver. Based on this, we propose a standard-compliant paging mechanism to manage the switching of tags between different states. So here are two questions: 1) What states does the tag need for effective transmission? 2) How to control the tag to switch between different states?

First, to meet the transmission requirements, the tag must be in a high-power active state to decode signals. This is due to the limited decoding capability in the low-power state, which prevents the tag from correctly decoding the signal. Second, the tag needs to enter a low-power sleep state when the UE goes to sleep for energy-saving and efficient transmission. To achieve the above two basic requirements, SyncScatter [31] designs a wake-up scheme to prevent the tag from being active all the time. However, for LTE backscatter, this method is not compatible with LTE, and the tag cannot determine PF and PO. We know that the paging mechanism is standard-compliant, and the LTE network only sends paging signals in the active state of the paging cycle. Therefore, we can also apply this mechanism to LTE backscatter. However, the standard paging mechanism cannot be used directly for LTE backscatter because an energy-constrained tag cannot identify PF and PO and decode paging messages, which requires high-power digitization and signal processing. To this end, we use the UE to forward the paging message to the tag, and the tag determines the PF and PO by identifying the paging. Paging identification requires high-accuracy sampling and decoding, and tags must be active for identification. In addition, although paging messages appear periodically, the surrounding LTE signal is continuous and will always excite the tag. the tag cannot first detect the arrival of a paging message in a low-power state and then identify it to reduce power consumption. As a result, the tag must remain in the high power state until the first paging message arrives to determine the PF and PO. To alleviate the energy waste caused by the tag being always active to identify paging before determining the PF and PO, we introduce a low-power wake-up strategy to indicate that the paging message is coming. During the wake-up process, the tag decodes the wake-up packet modulated using AM. Since simple wake-up does not require high-precision sampling, the tag can decode wake-up packets in a low-power wake-up state. As a result, the tag alternately switches between the wake-up state and the active state, starting from the initial static power state. When the tag does not find UE ID in the paging message, the tag enters the sleep state.

Fig. 5 shows the state switching of PScatter. The tag is initially in a static state. When a wake-up packet is detected, the tag enters the wake-up state and decodes the wake-up packet modulated by the AM. Once the tag is woken up, it enters the active state and identifies paging packets. When the paging information is identified, the tag determines the PF and PO, enters the periodic paging mode, and decodes the paging message. Since the UE ID is 110, its ID does not exist in the decoded message. Therefore, the tag switches from the active state to the sleep state and waits for the arrival of the next paging packet. After waiting for a paging cycle, the tag enters the active state again and continues to decode the paging message. The UE ID exists in this paging message, indicating that the UE is ready to receive signals, so the tag exits the paging mode. Since it supports a standard-compliant periodic paging mechanism, PScatter can switch

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between different states according to the UE's states.

C. Paging Forwarding

There are several difficulties with paging demodulation at the tag. First, paging messages from the base station are encapsulated in specific frames and subframes, and LTE frames are transmitted continuously. To receive a paging message, the tag must first locate a specific frame and subframe. This requires the tag to demodulate the Master Information Block (MIB) to obtain the System Frame Number (SFN) and align the subframes with downlink synchronization. However, only the determination of PF and PO requires power-starving signal processing and digitization, and the tag has to require a large number of computation resources and generate considerable power consumption. Secondly, it is impossible to judge whether there is a receiver in the area and whether the receiver is ready to receive signals based on the LTE signal from the base station alone. This requires the receiver to remind the tag of its own existence and transmission schedule.

1) AM paging forwarding: To solve the above issues, our solution is to forward the paging message using the UE. UE can decode the paging signal since it follows standard LTE. Hence, our key issue is to establish a reliable forwarding link from the UE to the tag. To enable low-power tags to decode forwarded paging messages, we need to decode packets using parameters that are easily measurable by the envelope detector, such as packet length and amplitude. Since PLM has a longer symbol duration than AM, it increases the decoding time, preventing the tag from processing the paging message within one cycle and introducing additional delay. This makes PLM unsuitable for latency-sensitive mobile applications due to the higher latency. To meet the reception requirement of various paging messages, we choose the AM to forward paging messages. The additional power consumption from the UE is minimal. This is due to the small size of the paging packet and the fact that the UE forwards paging in a low-duty cycle mode. Furthermore, the success rate of the UE forwarding paging is close to 100%, indicating high efficiency. Therefore, in comparison to other applications running on the UE, PScatter will only introduce a negligible increase in power consumption. For example, the tag transmits 100 bytes of text. We set a paging cycle to 320 ms and active duration to 20 ms. To transmit 100 B tag data, the total data transmission cycle required is $100 * 8/160 = 5$ cycles. As a result, 430 µs paging forwarding throughout 80 µs wake-up + 20,000 µs paging demodulation + 5*320,000 µs tag data transmission. In this case, the proportion of UE paging forwarding is $430/(80+20,000+5*320,000) = 2.65*10^{-4}$. Additionally, we use IC simulation to evaluate the digital core power of an additional function on UE for paging forwarding, which is 26 µW. Even if we consider the amplifier power, the power consumption of a 12dB gain RF amplifier is 240 µW [31]. It turns out that the extra power generated by the UE for paging forwarding is negligible since its power consumption is 0.07 µW.

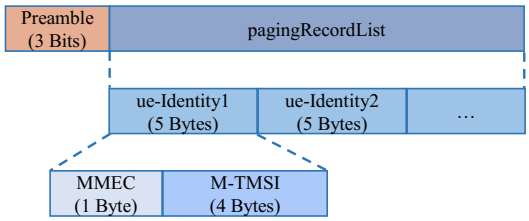

Fig. 6: The structure of the forwarded paging packet.

2) Forwarded paging structure and PF/PO determination: Fig. 6 shows the structure of the forwarded paging packet. The forwarded paging message contains the pagingRecordList of at least one *ue-Identity*, where the network uses SAE-Temporary Mobile Subscriber Identity (S-TMSI) as the UE identity, which consists of 1-byte Mobility Management Entity Code (MMEC) and 4-byte MME-Temporary Mobile Subscriber Identity (M-TMSI). To enable the tag to determine the PF and PO, we add a predefined preamble sequence to the header of the forwarded paging packet, which is an alternating sequence of 010. The tag can determine PF and PO by identifying this preamble. Therefore, the forwarded paging message is a sequence of more than 40 bits constructed from the preamble and pagingRecordList, where a low level represents bit 0 and a high level represents bit 1.

The main difference from standard paging reception is the determination of PF and PO. For standard paging reception, the UE must first synchronize the LTE signal and decode the MIB to locate the paging subframe and obtain SFN, which requires power-starving correlation and signal processing that energy-constrained tags cannot afford. To solve this issue, the tag takes the start of the identified paging packet as PF and PO. We identify paging messages by matching paging signals with a pre-stored preamble. Because the length of the preamble is fixed, we can determine PF and PO by going back to one preamble length from the identified time node.

D. Standard-incompatible UE Design

Paging forwarding using amplitude modulation is the sole approach tags can use to demodulate paging since passive LTE signal demodulation is highly challenging. Despite some backscatter techniques being able to demodulate LoRa and 802.11b signals, none can demodulate LTE signals based on OFDM [17], [34]. Currently, while integrating amplitude modulation functionality may not be the most optimal method, it serves as a feasible solution for LTE decoding. In the future, we will focus on exploring LTE demodulation methods that do not require UE relay. An alternative approach would be to adopt the WiFi-emulated ASK method from Interscatter [35] for implementing LTE-emulated ASK, given that both LTE and WiFi are OFDM signals.

IV. Putting It All Together

Fig. 7 shows the comparison of transmission between PScatter and LScatter [24]. PScatter tag is initially static in

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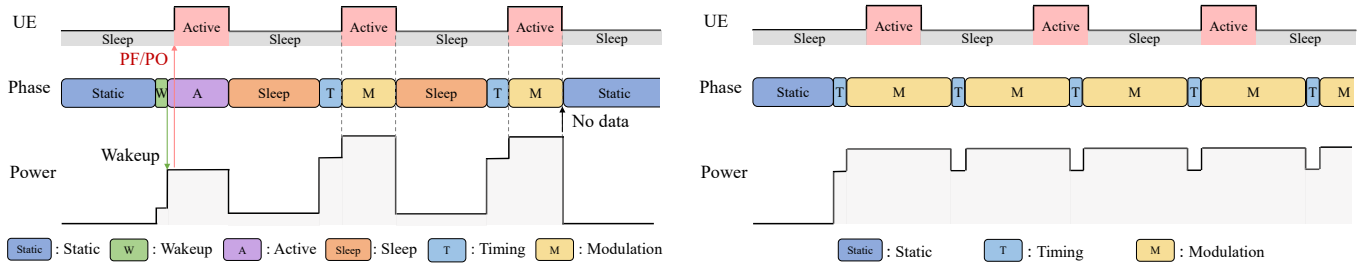
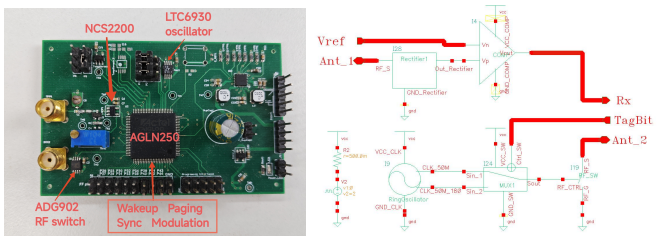


(a) PScatter. (b) LScatter.

Fig. 7: Comparison of PScatter and LScatter transmissions. In backscatter transmission, (a) PScatter tag maintains a consistent transmission schedule with the UE. (b) LScatter tags continuously backscatter transmissions. Thanks to the standard paging mode, PScatter not only saves more energy than LScatter but also transmits more efficiently.



(a) Verification prototype. (b) IC prototype.

Fig. 8: Two tag prototypes.

ultra-low-power Flash*Freeze mode. Once a wake-up packet is received indicating that a paging signal is imminent, it will switch from the wake-up phase to the active phase and decode the paging message forwarded by the UE in the periodic DRX mode. After confirming that the paging is sent to the UE, indicating that the UE is ready to receive signals, the tag modulates data when the UE is active according to the UE's sleep and wake schedule. When the tag does not intend to transmit data, it returns to the static state. Unlike PScatter, LScatter does not have a DRX mode that switches between active and sleep states and is always in the active state to backscatter signals. When the UE is dormant, it cannot receive backscattered signals, which causes LScatter transmissions to be ineffective and power-consuming. Therefore, PScatter can achieve energy-efficient and reliable transmission thanks to the duty-cycled paging mode.

Wake-up. As mentioned in Section III-B, before determining PF and PO, the tag is always in a high-power active state to identify paging, which leads to wasted energy. To reduce power consumption, we introduce a wake-up mechanism to indicate that the next packet is a paging packet. Similar to constructing the paging packet, we use AM to modulate the wake-up packet to reduce the delay. For the transmitter side, the transmitter sends a predefined sequence 10 to construct a wake-up pattern, and the duration of one symbol is 40 µs. On the receiver side, the tag acquires the baseband using a passive envelope detector and measures the high level during one symbol duration with a voltage comparator.

Timing. To determine the start of backscatter modulation, the tag must align with the LTE frame using timing. To reduce power consumption, we use quantization and downsampling. Tag quantizes envelopes with the comparator and reduces the sample depth to 1. Moreover, the tag reduces the sampling rate to 1 MHz, greatly reducing template storage. Nevertheless, insufficient sampling accuracy will cause PSS false positives. Hence, we utilize the periodicity of PSS, which has a period of 5 ms, store a single PSS template in the FPGA, and perform template matching every other cycle. When three consecutive periodic correlation results all exceed the correlation threshold, we consider the synchronization to be successful.

Modulation. After the tag is aligned with the LTE carrier, it starts to modulate data. Due to the complexity of the LTE frame structure, the control field and the data field are interleaved. To prevent control information, system parameters, and channel estimation from being destroyed, we reserve the control field, synchronization signal, and reference signal according to the LTE frame structure and only modulate data on the non-reference signal PDSCH symbols based on the codeword translation used in Hitchhike [29].

V. Implementation

We prototype PScatter using ultra-low-power FPGAs and USRPs. The implementation is detailed as follows.

LTE transceiver. We utilize the open-source software-defined radio LTE libraries OpenAirInterface [36] and srsRAN [37] with USRP B210 to run the eNodeB and UE.

Verification prototype. Our tag verification prototype for system performance is shown in Fig. 8a. It consists of two parts: the RF front-end, which transmits and receives signals, and an FPGA, which processes the baseband signals. The RF front end has an antenna connected to the envelope detector and voltage comparator. The other backscatter antenna is connected to an ADG902 RF switch. Our baseband processing, including wake-up, paging, synchronization, and backscatter modulation, is implemented on an Igloo nano AGLN250 FPGA, which has only 6,144 D-flip-flops.

IC prototype. We simulate an IC prototype using Cadence IC6.17 Virtuoso software and TSMC 0.18 µm CMOS process

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TABLE I: Power consumption comparison in end-to-end transmission.

	PScatter						LScatter+		
	Static	**wake-up**	**Active**	**Sleep**	**Timing**	**Modulation**	**Static**	**Timing**	**Modulation**
Verification Power (mW)	0.91	3.94	9.01	2.89	9.14	14.85	0.91	9.00	14.85
IC Power (μW)	10.47	45.61	59.04	14.35	89	149.35	10.47	18.98	149.35
Duration (ms)	N/A	0.08	20	8610	210	128	N/A	357	7382
Energy/bit (μJ/bit)	28.2 (Verification); 0.16 (IC)						110.2 (Verification); 1.08 (IC)		

TABLE II: Power comparison in one paging cycle.

	SyncScatter				LScatter		PScatter				Active LTE	
	Static	**wakeup**	**Active**	**Sleep**	**Static**	**Active**	**Static**	**wakeup**	**Active**	**Sleep**	**Sleep**	**Idle**
Power (mW)	0.91	3.75	9.00	9.00	0.91	9.00	0.91	3.94	9.01	3.00	4.90	63.10
Duration (ms)	N/A	0.68	640		N/A	640	N/A	0.08	1	639	N/A	640
Average power	8.99 mW				9.00 mW		3.01 mW				63.10 mW	

design kits. The IC prototype for a low-power design shown in Fig. 8b mainly consists of a detector including a rectifier and a comparator, an oscillator, a digital core with a small-size state machine, and an RF switch.

Experimental setup. The experimental setup for each part is as follows. UE initially receives paging on the downlink channel and subsequently forwards the paging on the uplink channel in band 14. Then, our tag reflects the LTE signal to a different channel by shifting the frequency, which effectively reduces self-interference. For paging, we set the duration of one symbol in the paging packet to be 10 μs. We use a 1 MHz clock to sample data in the active state and a low-frequency clock, 50 kHz, to count paging cycles in the sleep state. For wake-up, our tag uses a 200 kHz clock instead of 50 kHz to sample wake-up packets to improve decoding accuracy. SyncScatter tag uses a 50 kHz clock, and the L0 and L1 in the PLM package are set to 140 μs and 340 μs, respectively. Moreover, the packet interval is 200 μs. For timing and modulation, we use a 1 MHz clock to sample carrier signals and modulate the tag data. In addition, the tag uses a 25 Mhz clock for frequency shift.

Comparing with baseline approaches. We mainly compare PScatter against two state-of-the-art backscatter schemes, LScatter [24], and SyncScatter [31]. LScatter is a typical LTE backscatter scheme that does not save energy from the core logic. SyncScatter itself has a duty cycle mode. Since it cannot determine PF/PO, the SyncScatter tag cannot accurately align UE's transmission schedule. Also, we compare the power with Quectel's LTE module, EC200N-CN, which is active LTE. Here, we refer to the hardware design manual of EC200N-CN. For end-to-end transmission, we compare PScatter with LScatter and introduce LScatter+, which combines the synchronization of LScatter and the BPSK modulation of PScatter.

VI. Evaluation

A. Energy Efficiency

We evaluate the energy efficiency of PScatter, LScatter, and LScatter+ in end-to-end transmission as shown in Table I. In our experiment, we set the tag data size to 1 Kbit and the active duration in one paging cycle to 20 ms. We observe that implementing a standard-compatible paging mode allows the LTE tag to dynamically adjust states based on network traffic. This adaptation enhances energy efficiency and promotes efficient data transmission. The energy efficiencies of PScatter for verification and IC are 28.2 μJ/bit and 0.16 μJ/bit, which achieves 3.9x and 6.7x better than LScatter+, respectively. In addition, as the active duration decreases, the ratio of the energy efficiency of LScatter+ to PScatter increases. The reason is that different active durations bring different transmission durations and paging cycles; the shorter the duration, the longer the total transmission time, and the higher the total energy. This will waste more energy for LScatter+, which does not support paging. Even if we choose the maximum active duration (200 ms), the energy efficiencies of PScatter and LScatter+ for IC are 0.05 μJ/bit and 0.17 μJ/bit, respectively. Compared to LScatter, which disrupts the LTE structure with its modulation at the sub-symbol level, standard LTE UEs are unable to correctly decode the backscatter signal. Note that LTE networks transmit paging messages with a low-duty cycle. However, LScatter UEs cannot demodulate tag data carried on blank subframes. Using LTE UEs without the low-duty cycle mode leads to huge energy waste in UEs. Therefore, we chose not to use LScatter with such LTE UEs.

B. Power Consumption

We output different state control signals to indicate the tag state. For example, when the control signal 'active' is 1, it means that the tag enters the active state of paging. According to these signals, we measure the current of the FPGA in the corresponding state and calculate the power in this state by the formula $P = I * V$. Finally, we use the measured power and duration in different states to estimate the average power. In addition, since the static power of different methods is the same and the time is uncertain, this power consumption is not included in the total power consumption.

Power comparison in one paging cycle. We evaluate the power comparison of different methods in one paging cycle. As shown in Table II, the average power consumption of PScatter during one paging cycle is the lowest at 3.01 mW, which is 21.0x lower than active LTE EC200N-CN, and 2.99x lower than SyncScatter and LScatter, respectively. The reason is that PScatter introduces a periodic paging mode, allowing it to demodulate paging messages and enter sleep mode at appropriate times. In contrast, SyncScatter's wake-up

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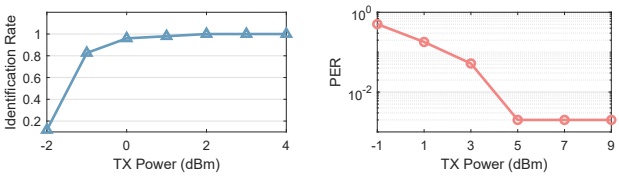
TABLE III: PScatter power in different states.

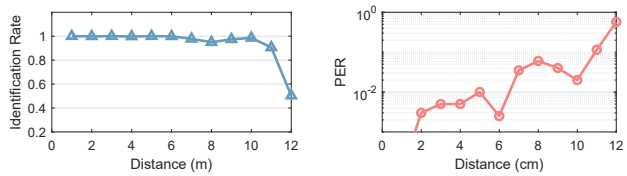
State	Verification Power (mW)						IC Power (μW)					
	FPGA	Comparator	Oscillators			Total	Digital core	Comparator	Oscillators			Total
			50 kHz	200 kHz	1MHz				50 kHz	200 kHz	1MHz	
Static	0.66	0.18	0.07	N/A	N/A	**0.91**	0.24	10	0.23	N/A	N/A	**10.47**
wake-up	3.50	0.18	N/A	0.26	N/A	**3.94**	34.75	10	N/A	0.86	N/A	**45.61**
Active	8.76	0.18	0.07	N/A	0[1]	**9.01**	45.06	10	N/A	N/A	3.98	**59.04**
Sleep	2.75	0.18	0.07	N/A	N/A	**3.00**	4.12	10	0.23	N/A	N/A	**14.35**

[1] Actual measured value is 0.36, included in FPGA power 8.76 mW.



(a) Paging identification rate. (b) PER.

Fig. 9: Paging demodulation across TX powers.



(a) Paging identification rate. (b) PER.

Fig. 10: Paging demodulation across distances.

is incompatible with LTE, making it unable to determine PF and PO, thus remaining active after wake-up to receive paging messages. LScatter lacks a duty cycle mode and remains active continuously. Active LTE involves energy-intensive signal processing procedures. Note that SyncScatter's wake-up duration is 680 μs instead of the same 80 μs as PScatter because the symbol duration using PLM is longer than AM.

Power in different states. We also evaluate the power of PScatter in different states. Table III demonstrates PScatter's efficient energy savings achieved through a duty cycle paging mode. Specifically, PScatter's IC power in static, wake-up, active, and sleep states are 10.47 μW, 45.61 μW, 59.04 μW, and 14.35 μW, respectively. The power consumption in sleep mode is 4.1x lower than in active mode. Additionally, for verification power, the FPGA power in active mode, including the 1 MHz oscillator, is 3.2x higher than in sleep mode.

Impact of different paging cycles. Moreover, we evaluate the average power comparison of PScatter and SyncScatter under different paging cycles. As shown in Fig. 11, we observe that under any paging cycle, the average power of PScatter is lower than that of SyncScatter. Specifically, for 32 rf and 256 rf paging cycles, PScatter consumes average power that is 2.7x and 3.1x lower than SyncScatter, respectively. Additionally, we observe that SyncScatter's average power remains relatively stable across different paging cycles, with the maximum difference not exceeding 10 μW. This stability arises because varying paging cycles primarily impact the duration SyncScatter spends in its active state, where the active duration significantly outweighs the wake-up duration within a single paging cycle. Consequently, the average power of SyncScatter closely approximates its active state power. In contrast, PScatter shows a gradual decrease in average power as the paging cycle increases, with the maximum power difference being 0.4 mW. Unlike SyncScatter, changes in the paging cycle primarily affect the sleep duration of PScatter, as its active duration remains constant. Furthermore, due to the much longer sleep duration compared to the active duration within one paging cycle, the change in average power for PScatter is not prominently noticeable.

C. Paging Reception

1) Impact of transmission power:

Paging identification rate. As shown in Fig. 9a, we evaluate the paging identification rate at different transmission powers of UE. We observe that the maximum identification rate of PScatter is 100%. When the transmission power is not lower than 0 dBm, the identification rate reaches 0.95. However, when the transmission power is lower than -1 dBm, the identification rate is very low, close to 0. The reason is that the received signal strength is weak, and the Signal-to-Noise Ratio (SNR) is low, which causes the data to be distorted.

PER of paging messages. We also evaluate the packet error rate (PER) of paging messages over transmission powers. As shown in Fig. 9b, when the transmission power reaches 5 dBm, PER is 0.2%, which is low enough for paging demodulation. However, when the transmission power is below 1 dBm, the PER is higher than 10%. Especially when the transmission power is not higher than -1 dBm, the PER is up to 50%, which is much higher than the error rate of paging identification. There are two reasons for this notable gap. Firstly, the tag must accurately identify the paging information before demodulation. Secondly, the paging message consists of up to 40 bits, which far exceeds the duration of its preamble.

2) Impact of the distance between UE and tag:

To evaluate paging reception across distances between the UE and tag, we use an ADC, AD9235, to sample the paging signal. In our experiment, we adopt the moving average method to reduce the impact of sudden noise interference. At the start of each packet, we dynamically determine the quantization threshold based on the average signal strength measured over a 10 μs period. Subsequently, we use this threshold to quantize the signal. Besides, since the transmission power of the USRP is significantly lower than that of a typical UE

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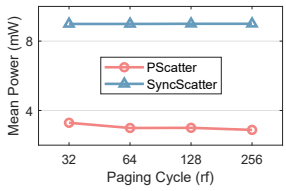

Fig. 11: Impact of paging cycle.

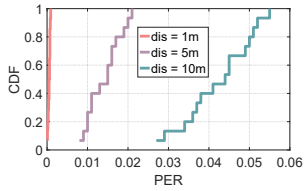

Fig. 12: Paging over commodity eNB.

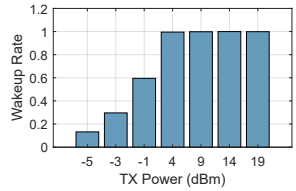

Fig. 13: Wake-up rate at different tx powers.

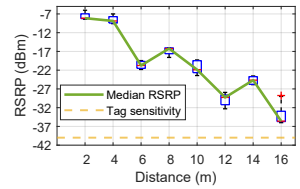

Fig. 14: Distance between eNB and tag.

(which can transmit up to 23 dBm), we simulate a real UE by boosting the UE transmission power to 20 dBm using an SPF5189Z power amplifier along with a 6 dBi antenna gain. Referring to SyncScatter [31], we also minimize the detector bandwidth while ensuring high demodulation quality to enhance tag sensitivity. Its sensitivity is about -25 dBm.

We evaluate the paging identification rate and PER over different downlink distances. Fig. 10a shows that the maximum paging identification rate is 100%. when the distance is within 10 m, the paging identification rate is above 95%. Fig. 10b shows that the PER keeps increasing with distance. When the distance is within 5 m, the PER does not exceed 1%. When the distance reaches 11 m, the PER is close to 10%.

3) Paging demodulation over commodity eNodeB: To verify that PScatter has the ability to communicate with commodity eNB. We use the Cellular-Z [38] app to search for the LTE signal from the commodity eNB deployed by the operator and set the dl_earfcn of the SDR srsUE to 1850 according to actual earfcn. Fig. 12 shows that when the distance between the UE and the tag is 1 m, the 80th percentile of PER for forwarded paging demodulation is 0.07%, and when the distance is 10 m, the 80th percentile of PER is 5%.

D. Wake-up Rate

We evaluate the wake-up rate at different transmission powers. As shown in Fig. 13, we observe that the maximum wake-up rate is almost 100%. When the transmission power is not lower than 4 dBm, the wake-up rate exceeds 0.99 without significant change because when the transmission power reaches 4 dBm, the SNR meets the wake-up requirement.

E. Distance between eNB and Tag.

Finally, we evaluate the distance between eNB and tag. We set the transmission power of the eNB to 30 dBm (the typical transmission power is 43 dBm). Since backscatter is not required to demodulate the signal from the eNB, we use a low-bandwidth envelope detector. As a result, the RSRP sensitivity of the tag is -40 dBm. Fig. 14 shows that the maximum distance within tag sensitivity is 16 m. For a typical real LTE base station with a transmission power of 43 dBm, the maximum predicted range is 19 m.

VII. Related Work

Inspired by prior ambient backscatter works such as WiFi [39]–[41], Bluetooth [42], [43], LoRa [34], [44] and mmWave [45], [46], PScatter proposes energy-efficient paging for LTE backscatter. However, none of the past LTE backscatter works have achieved efficient energy saving. PScatter is the first LTE backscatter system to reduce power consumption from the core control, which is mainly inspired by the following works.

For active radios, LTE periodically controls UE switching states using the DRX mechanism to reduce power consumption. [47] proposes an LTE smartphone power model and measures DRX power consumption from the system level. [28] derives the first comprehensive power model for commercial LTE networks to characterize LTE power usage. Wake-on-Wireless [27] proposes to reduce idle power to increase the battery life of the device. Mobile IoT technologies, LTE-M and NB-IoT, introduce Extended idle Mode DRX (eDRX) and Power Saving Mode (PSM) to save energy, and WiFi introduces the PS-NULL-Poll mechanism to save power. For backscatter radios, most systems reduce power consumption by designing low-power circuits [17], [48]–[50]. However, few backscatter systems save energy from logic control [51]–[53]. LScatter [24], a typical backscatter system, is always active for communication. Therefore, the existing energy-saving solutions for backscatter cannot be used for LTE backscatter.

VIII. Conclusion

We have presented PScatter, an efficient LTE backscatter design that enhances energy efficiency and ensures reliable transmission by dynamically managing backscatter devices according to the UE's low-duty cycle mode. Employing standard paging, we control tag states and forward paging messages, enabling low-power tags to efficiently decode messages using the UE. We have built hardware prototypes and conducted extensive experiments to verify the feasibility and efficacy. We believe LTE backscatter, with support for standard paging, provides reliable transmission and prolongs device lifespan, significantly enhancing its potential for large-scale deployment and applications that are battery-free.

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